

PROJECT DEVELOPMENT PHASE 1

Data Preprocessing Steps

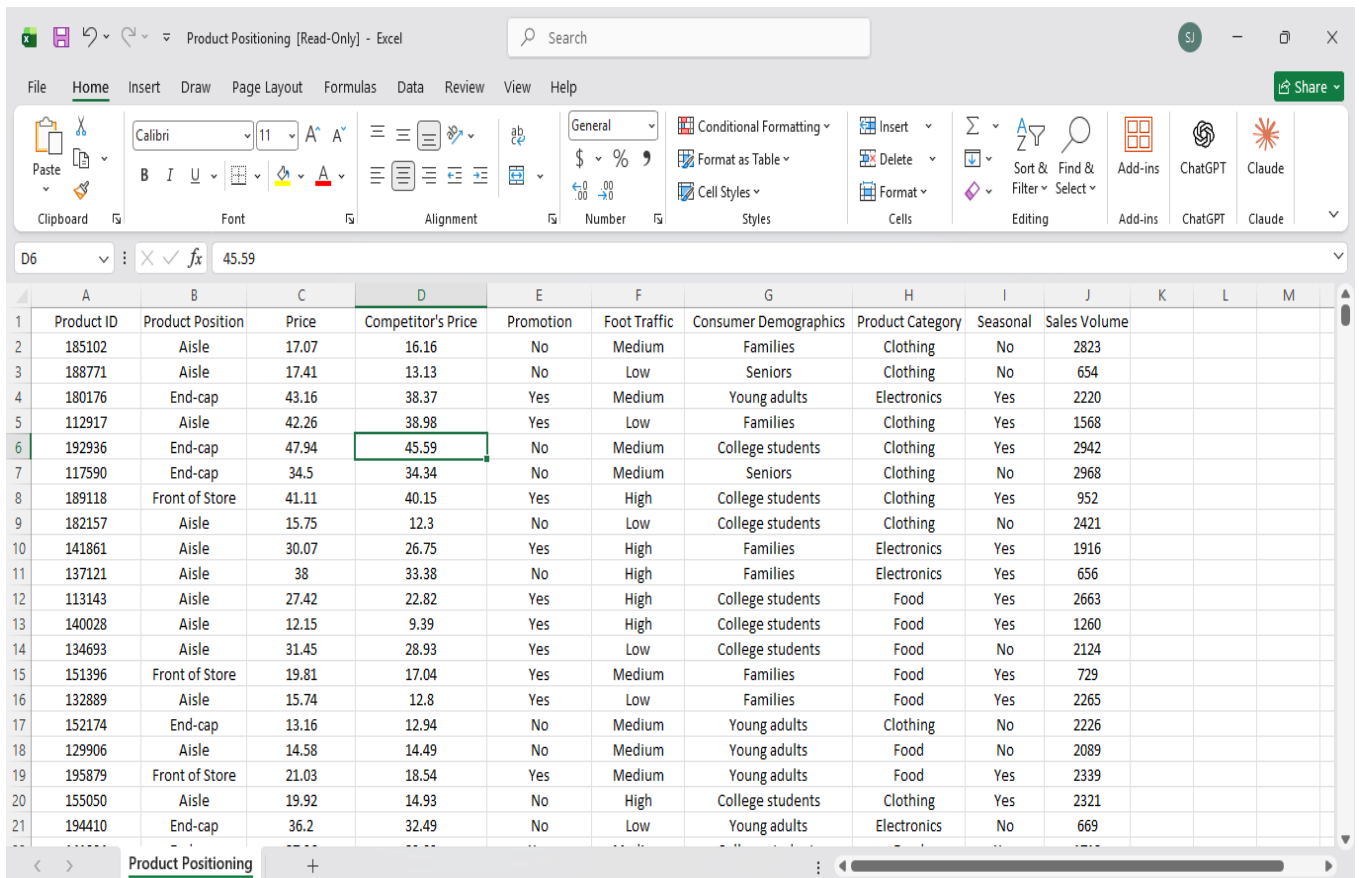
Step 1: Dataset Collection

The dataset used in this project, **"Impact of Product Positioning on Sales,"** was downloaded from Kaggle in CSV format. It contains retail sales data related to product positioning, product pricing, competitor pricing, promotional activities, customer demographics, seasonal trends, foot traffic, and sales volume. This dataset served as the primary source for developing the Business Intelligence dashboard.

Dataset Source:

<https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales>

 **Figure 1.1: Product Positioning Dataset in Microsoft Excel**



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Product ID	Product Position	Price	Competitor's Price	Promotion	Foot Traffic	Consumer Demographics	Product Category	Seasonal	Sales Volume			
2	185102	Aisle	17.07	16.16	No	Medium	Families	Clothing	No	2823			
3	188771	Aisle	17.41	13.13	No	Low	Seniors	Clothing	No	654			
4	180176	End-cap	43.16	38.37	Yes	Medium	Young adults	Electronics	Yes	2220			
5	112917	Aisle	42.26	38.98	Yes	Low	Families	Clothing	Yes	1568			
6	192936	End-cap	47.94	45.59	No	Medium	College students	Clothing	Yes	2942			
7	117590	End-cap	34.5	34.34	No	Medium	Seniors	Clothing	No	2968			
8	189118	Front of Store	41.11	40.15	Yes	High	College students	Clothing	Yes	952			
9	182157	Aisle	15.75	12.3	No	Low	College students	Clothing	No	2421			
10	141861	Aisle	30.07	26.75	Yes	High	Families	Electronics	Yes	1916			
11	137121	Aisle	38	33.38	No	High	Families	Electronics	Yes	656			
12	113143	Aisle	27.42	22.82	Yes	High	College students	Food	Yes	2663			
13	140028	Aisle	12.15	9.39	Yes	High	College students	Food	Yes	1260			
14	134693	Aisle	31.45	28.93	Yes	Low	College students	Food	No	2124			
15	151396	Front of Store	19.81	17.04	Yes	Medium	Families	Food	Yes	729			
16	132889	Aisle	15.74	12.8	Yes	Low	Families	Food	Yes	2265			
17	152174	End-cap	13.16	12.94	No	Medium	Young adults	Clothing	No	2226			
18	129906	Aisle	14.58	14.49	No	Medium	Young adults	Food	No	2089			
19	195879	Front of Store	21.03	18.54	Yes	Medium	Young adults	Food	Yes	2339			
20	155050	Aisle	19.92	14.93	No	High	College students	Clothing	Yes	2321			
21	194410	End-cap	36.2	32.49	No	Low	Young adults	Electronics	No	669			

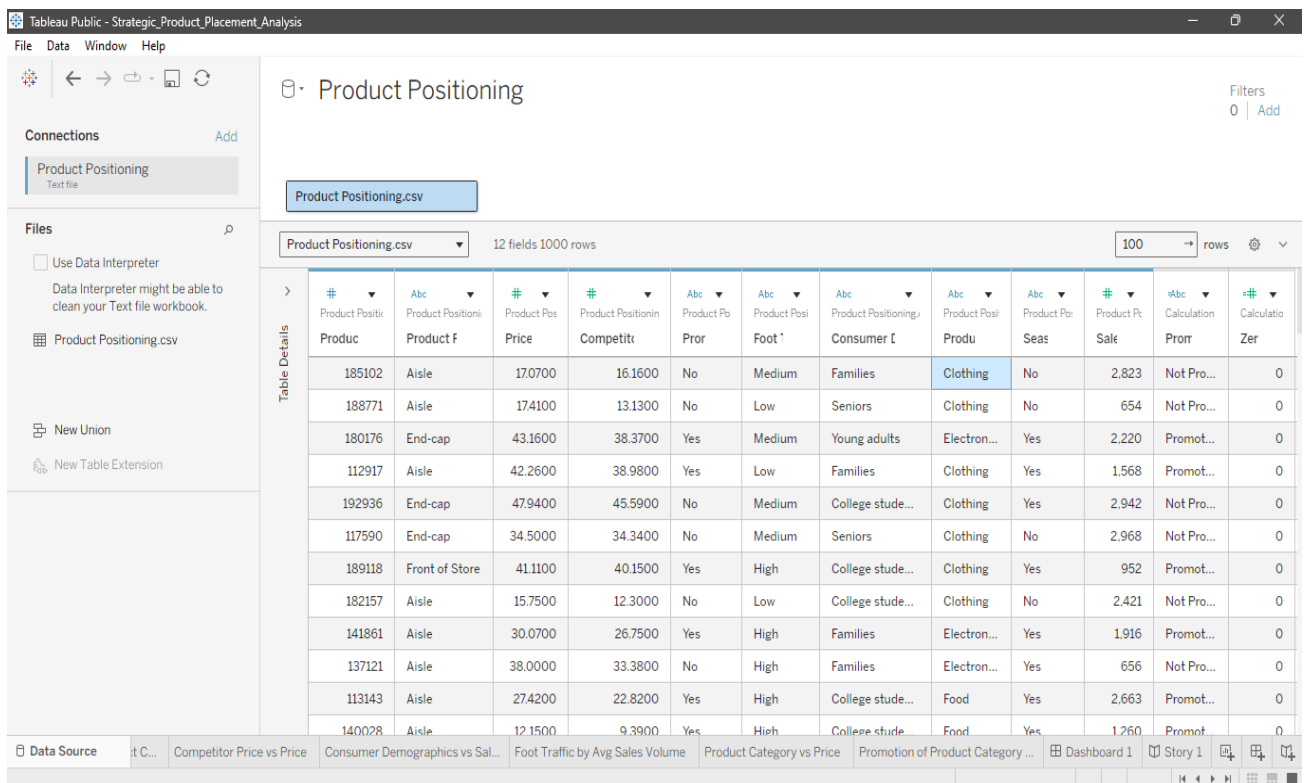
Step 2: Opening the Dataset

The downloaded CSV file was opened in Microsoft Excel to understand the structure of the dataset and verify the available fields before importing it into Tableau Public.

Step 3: Importing Dataset into Tableau Public

The CSV file was imported into Tableau Public using the **Text File** connection. Tableau successfully loaded the dataset containing **1000 records** and displayed all the fields required for analysis.

 **Figure 1.2: Dataset Imported into Tableau Public**



The screenshot shows the Tableau Public interface with the 'Product Positioning' dataset loaded. The main view displays a table with 12 fields and 1000 rows. The fields are: Product ID, Product Name, Price, Competitor Price, Product Category, Product Type, Consumer Type, Product Location, Season, Sales, Promotion, and Zero. The table is filtered to show 100 rows.

Product ID	Product Name	Price	Competitor Price	Product Category	Product Type	Consumer Type	Product Location	Season	Sales	Promotion	Zero
185102	Aisle	170700	16.1600	No	Medium	Families	Clothing	No	2.823	Not Pro...	0
188771	Aisle	174100	13.1300	No	Low	Seniors	Clothing	No	654	Not Pro...	0
180176	End-cap	43.1600	38.3700	Yes	Medium	Young adults	Electron...	Yes	2.220	Promot...	0
112917	Aisle	42.2600	38.9800	Yes	Low	Families	Clothing	Yes	1.568	Promot...	0
192936	End-cap	47.9400	45.5900	No	Medium	College stude...	Clothing	Yes	2.942	Not Pro...	0
117590	End-cap	34.5000	34.3400	No	Medium	Seniors	Clothing	No	2.968	Not Pro...	0
189118	Front of Store	41.1100	40.1500	Yes	High	College stude...	Clothing	Yes	952	Promot...	0
182157	Aisle	15.7500	12.3000	No	Low	College stude...	Clothing	No	2.421	Not Pro...	0
141861	Aisle	30.0700	26.7500	Yes	High	Families	Electron...	Yes	1.916	Promot...	0
137121	Aisle	38.0000	33.3800	No	High	Families	Electron...	Yes	656	Not Pro...	0
113143	Aisle	27.4200	22.8200	Yes	High	College stude...	Food	Yes	2.663	Promot...	0
140028	Aisle	12.1500	9.3900	Yes	High	College stude...	Food	Yes	1.260	Promot...	0

Step 4: Renaming the Field

To improve readability and maintain consistency throughout the dashboard, the field "Competitor's Price" was renamed to "Competitor Price".

Step 5: Verifying Data Types

After importing the dataset, the data types of the required fields were verified and modified where necessary to ensure proper visualization and analysis in Tableau Public.

Step 6: Creating Calculated Fields

The required calculated fields were created in Tableau Public to support dashboard development and improve data visualization.

Step 7: Dataset Ready for Visualization

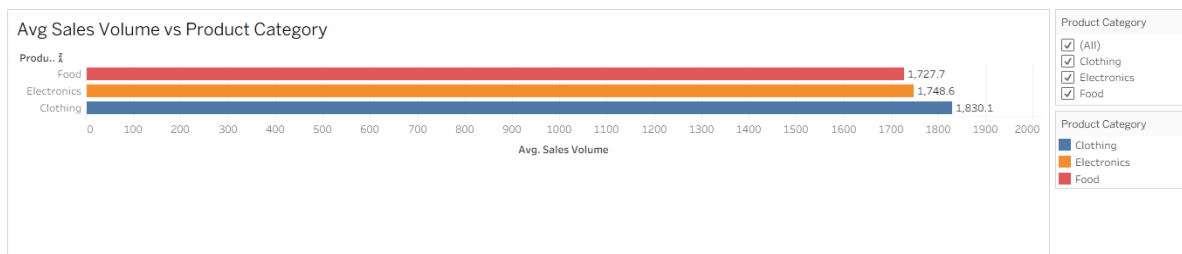
After importing the dataset, renaming the required field, verifying the data types, and creating the calculated fields, the dataset was ready for creating worksheets, dashboards, and Tableau Stories.

Business Questions with Visualization

Question 1. Which Product Category Has the Highest Average Sales Volume?

Visualization Used: Bar Chart

 **Figure 1.3: Average Sales Volume by Product Category**



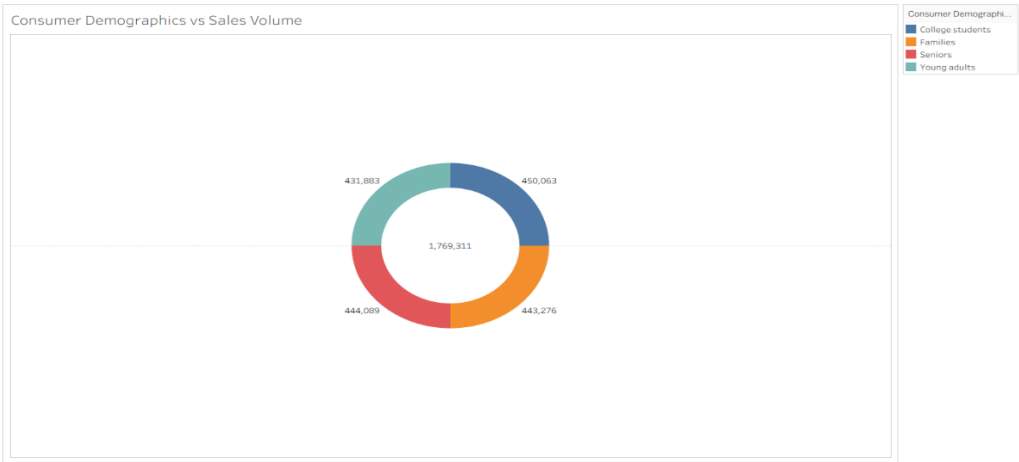
Business Insight:

This visualization compares the average sales volume of Clothing, Electronics, and Food categories. It helps identify the best-performing product category and supports better inventory and product placement decisions.

Question 2. How Do Consumer Demographics Influence Sales Volume?

Visualization Used: Donut Chart

 Figure 1.4: Consumer Demographics vs Sales Volume



Business Insight:

The visualization shows the contribution of different customer groups such as College Students, Families, Seniors, and Young Adults to the overall sales volume. This information helps businesses understand customer preferences and develop targeted marketing strategies.

Question 3. What is the Impact of Promotions on Product Price and Sales Volume?

Visualization Used: Text Table

 Figure 1.5: Promotion of Product Category on Price and Sales Volume

Promotion of Product Category on Price and Sales Volume			
Pro..	2	Product Category	
No			Avg. Price
			Avg. Sales Volume
No		Clothing	27
		Electronics	27
		Food	28
Yes		Clothing	29
		Electronics	29
		Food	29

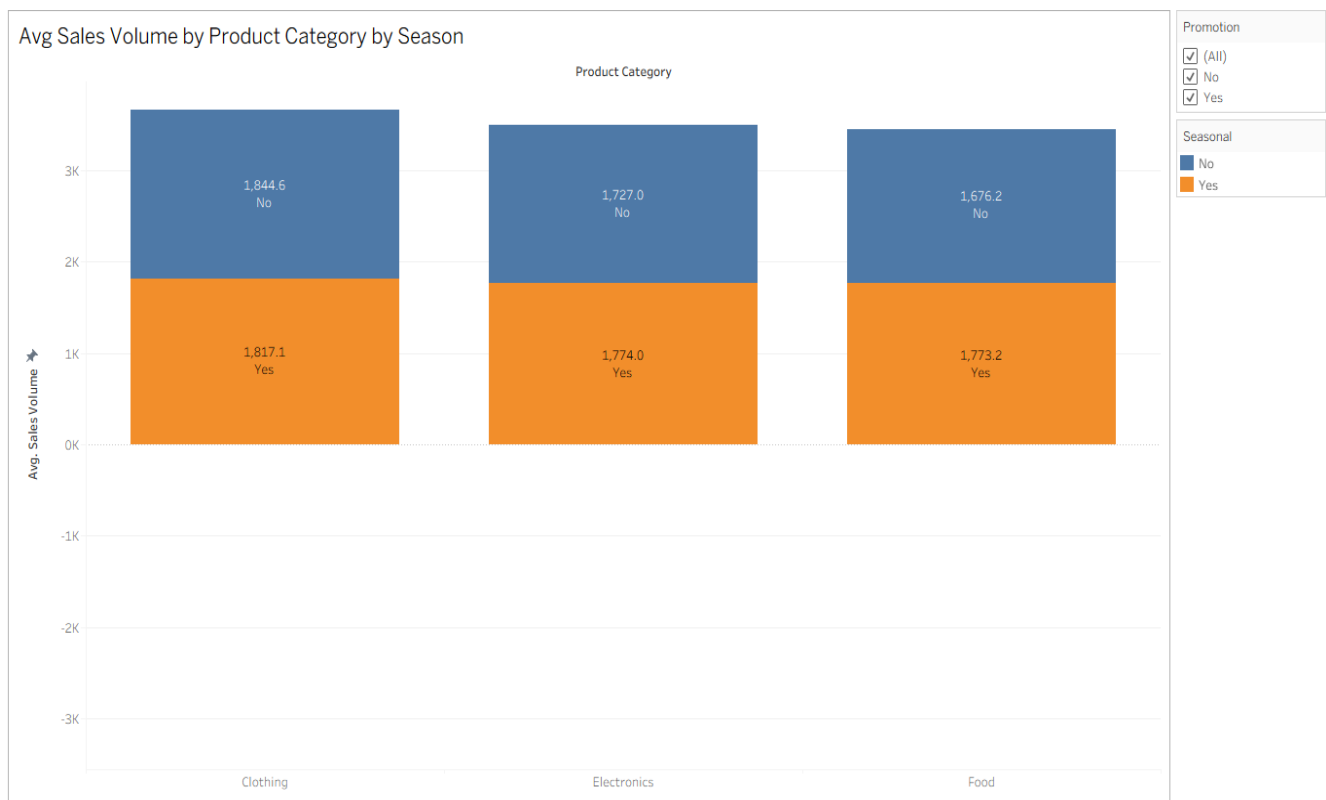
Business Insight:

The visualization compares promoted and non-promoted products across different product categories, helping businesses evaluate the effectiveness of promotional campaigns.

Question 4. How Does Seasonal Demand Affect Average Sales Volume Across Product Categories?

Visualization Used: Stacked Bar Chart

 **Figure 1.6: Average Sales Volume by Product Category by Seasonal**



Business Insight:

The chart illustrates how seasonal demand influences average sales volume across different product categories, enabling retailers to plan seasonal promotions and inventory more effectively.

Question 5. How Do Competitor Prices Compare with Product Prices?

Visualization Used: Grouped Bar Chart

📷 **Figure 1.7: Competitor Price vs Product Price**



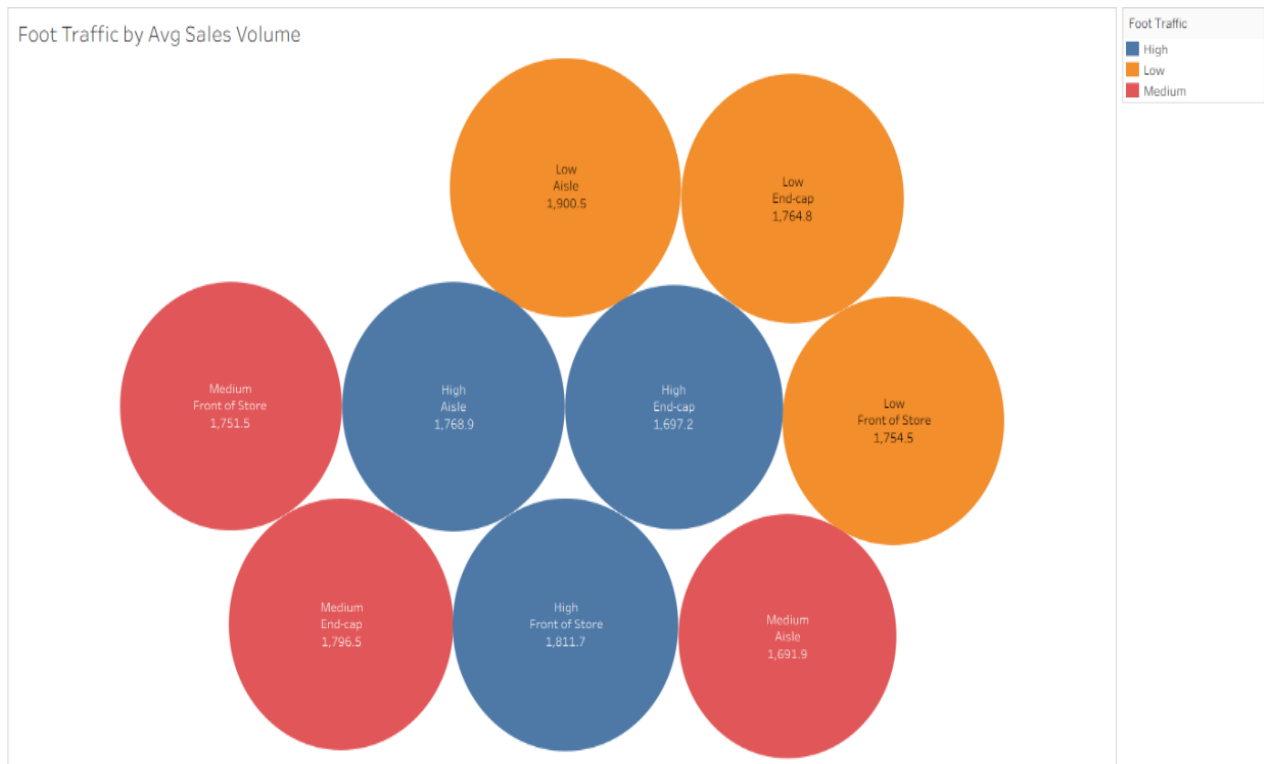
Business Insight:

The visualization compares product prices with competitor prices, allowing businesses to evaluate their pricing strategy and maintain competitive market positioning.

Question 6. How Does Foot Traffic Influence Average Sales Volume?

Visualization Used: Bubble Chart

 **Figure 1.8: Foot Traffic by Average Sales Volume**



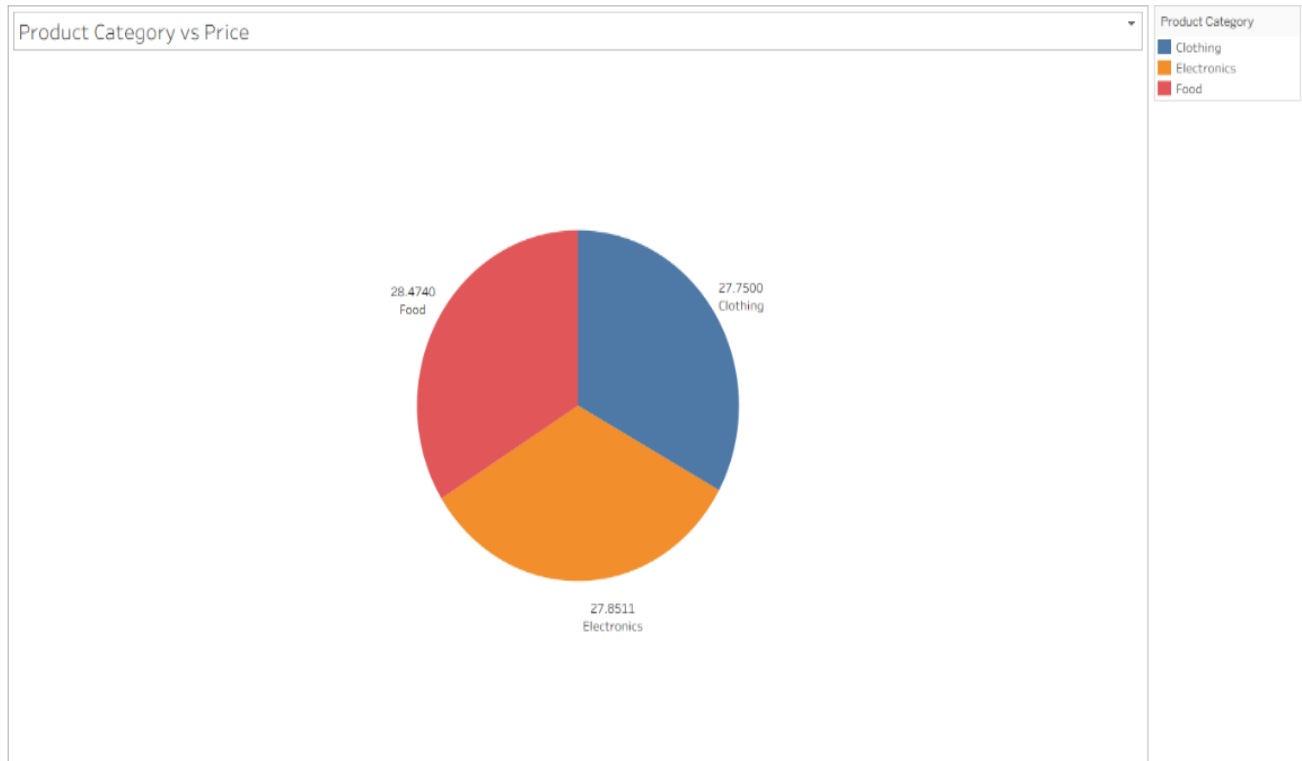
Business Insight:

The bubble chart illustrates the relationship between customer foot traffic levels and average sales volume. It helps identify how store traffic affects product sales performance.

Question 7. What is the Average Price Distribution Across Product Categories?

Visualization Used: Pie Chart

 **Figure 1.9: Product Category vs Price**



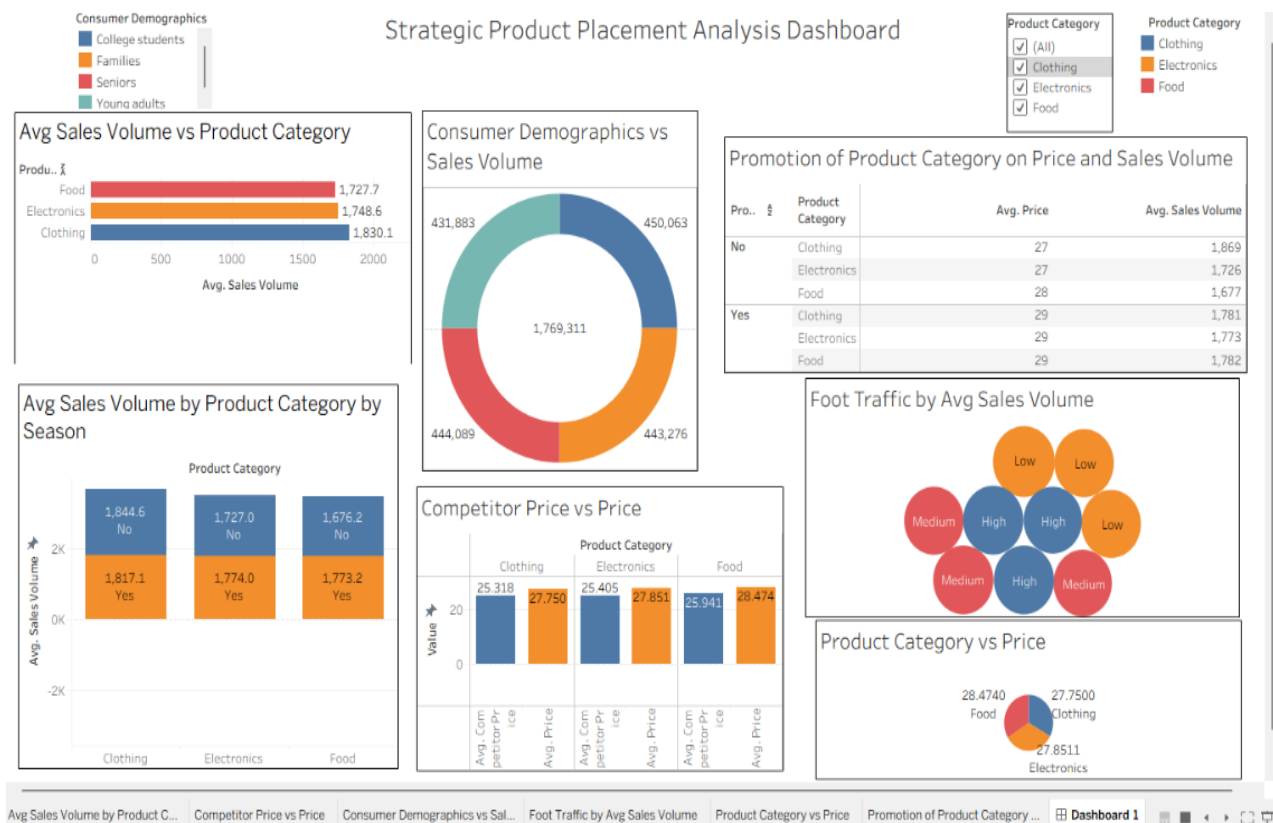
Business Insight:

The pie chart represents the average price distribution among Clothing, Electronics, and Food categories, providing an overview of pricing patterns across product categories.

Question 8. What Overall Business Insights Can Be Obtained from the Dashboard?

Visualization Used: Interactive Tableau Dashboard

Figure 1.10: Strategic Product Placement Dashboard



Business Insight:

The dashboard integrates all visualizations into a single interactive interface. It enables business managers to analyze product performance, customer demographics, pricing strategies, promotional effectiveness, seasonal trends, and foot traffic together. These insights support informed decision-making, optimize product placement strategies, and improve overall sales performance.

Summary

The preprocessing phase involved collecting the dataset, importing it into Tableau Public, renaming the required field, verifying the data types, and creating calculated fields for visualization. The processed dataset was then used to develop interactive worksheets, dashboards, and Tableau Stories. The visualizations answered key business questions related to product category performance, customer demographics, promotional effectiveness, pricing strategies, seasonal demand, and foot traffic. These insights help businesses make informed decisions and improve product placement strategies.

Figures Used

Figure	Description
--------	-------------

Figure 1.1	Dataset Opened in Microsoft Excel
------------	-----------------------------------

Figure 1.2	Dataset Imported into Tableau Public
------------	--------------------------------------

Figure 1.3	Average Sales Volume by Product Category
------------	--

Figure 1.4	Consumer Demographics vs Sales Volume
------------	---------------------------------------

Figure 1.5	Promotion of Product Category on Price and Sales Volume
------------	---

Figure 1.6	Average Sales Volume by Product Category by Seasonal
------------	--

Figure 1.7	Competitor Price vs Product Price
------------	-----------------------------------

Figure 1.8	Foot Traffic by Average Sales Volume
------------	--------------------------------------

Figure 1.9	Product Category vs Price
------------	---------------------------

Figure 1.10	Strategic Product Placement Dashboard
-------------	---------------------------------------